The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF).

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 Besnard-Hunter Deductive Argumentation and Defeasible Self-Repair

We formalize the machine-to-machine deconfliction and self-repair mechanism using Philippe Besnard and Anthony Hunter's deductive argumentation framework, replacing monotonic First-Order Logic (FOL) to handle localized theory anomalies.

3.1 The Support-Claim Argument Tuple

Let $\mathcal{K}_i$ represent the localized, non-monotonic knowledge base of an agent $a_i \in \mathcal{A}$. An argument $\mathcal{A}_k$ is formalized as a structural tuple:

$$\mathcal{A}_k = \langle \Phi, \alpha \rangle \tag{2}$$

where $\Phi \subseteq \mathcal{K}_i$ represents the *Support* (a minimal, logically consistent subset of local premises) and α represents the *Claim* (the derived behavioral heuristic or decision tree split command). The support must satisfy the following consistency criteria:

$$\Phi \nvdash \bot \quad \text{and} \quad \Phi \vdash \alpha \tag{3}$$

3.2 Topological Undercutters and Attack Primitives

When a source agent maps an anomalous theory, a target agent constructs a logical Counter-Argument designed to attack the consistency of the source's support. An argument $\mathcal{A}_2 = \langle \Phi_2, \alpha_2 \rangle$ is an *Undercutter* for $\mathcal{A}_1 = \langle \Phi_1, \alpha_1 \rangle$ if the claim α_2 directly negates a premise within the source support set:

$$\alpha_2 \equiv \neg\phi \quad \text{where} \quad \phi \in \Phi_1 \tag{4}$$

Upon receiving the undercutter payload over the Model Context Protocol (MCP) gateway, the claim is classified as defeated. This triggers an immediate 5-step inductive replacement loop, forcing the source agent to execute local back-propagation pruning across its decision forest $\mathcal{F}_1$ to realign the abstract tree topology with the master invariants ($\top$) without causing a global system halt.